DATA ANALYTICS PROJECT REPORT

Customer Churn & Retention Analytics

Understanding Customer Behavior, Churn Patterns & Retention Opportunities

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1. Abstract

This project analyses customer churn in a dataset of 8,000 customers. Of these, 2,505 customers churned, which gives an overall churn rate of 31.31%. Churn was compared across subscription type, location, account age, monthly spending, satisfaction score and support calls, and the correlation between each numeric column and churn was calculated.

The main result is that churn is spread broadly across the customer base. Group-level churn rates mostly sit between about 29% and 34%, and every correlation with churn is between roughly -0.02 and +0.02. No single variable stands out as a strong standalone churn driver. A few segments, such as Florida and accounts that are 25 to 48 months old, show slightly higher churn and could be a starting point for further investigation. The analysis is descriptive only and does not claim that any variable causes churn.

The results are presented in a Jupyter notebook and in an interactive dashboard built with HTML, JavaScript and Plotly.js.

2. Introduction

Churn means that a customer stops using a service. For a subscription business, every lost customer is lost recurring revenue, and winning a replacement costs time and money. Understanding which customers leave is a common first step before planning any retention activity.

This project applies basic exploratory data analysis and dashboard design to a customer dataset. The work was done with Python (pandas, NumPy, Matplotlib) for the analysis and with HTML, CSS, JavaScript and Plotly.js for the dashboard. No prediction model was built.

3. Business Problem

A business wants to reduce churn but has limited time and budget for retention campaigns. Before spending on them it needs answers to a few basic questions: how many customers churn, which groups churn more than others, and whether one factor (for example low satisfaction or many support calls) clearly separates churned customers from retained ones. If no such factor exists, a single blanket campaign is unlikely to work well and a more segmented approach is needed.

4. Objectives

- Count retained and churned customers and calculate the overall churn rate.
- Compare churn across subscription type, location, account age, spending, satisfaction score and support calls.
- Check whether any numeric variable has a strong relationship with churn.
- Build an interactive dashboard that recalculates when filters are applied.
- Suggest retention actions that are based only on the observed patterns.

5. Dataset Description

The analysis uses one file, **train.csv**, with 8,000 rows and 17 columns. The target column is **Churn** (0 = retained, 1 = churned). The table below lists each column with its type and the range or categories found in the data.

Column	Type	Values in the data
Customer_ID	Number	1001 to 9000 (identifier)
Age	Number	18 to 69
Gender	Category	Female, Male
Location	Category	California, Florida, Illinois, New York, Texas
Subscription_Type	Category	Basic, Enterprise, Premium
Account_Age_Months	Number	1 to 59
Monthly_Spending	Number	10.09 to 199.94
Total_Usage_Hours	Number	10 to 499
Support_Calls	Number	0 to 9
Late_Payments	Number	0 to 4
Streaming_Usage	Number	0 to 99
Discount_Used	Number	0 to 99
Satisfaction_Score	Number	1 to 10
Last_Interaction_Type	Category	Negative, Neutral, Positive
Complaint_Tickets	Number	0 to 4
Promo_Opted_In	Number	0 to 1
Churn	Number	0 to 1

The dataset was provided for this project. Its original source is not documented in this report. The test file was not used.

6. Data Quality

Before any analysis the file was checked for basic quality problems.

Check	Result
Rows	8,000
Columns	17
Missing values	0
Duplicate rows	0
Duplicate Customer_IDs	0
Values in Churn	0 and 1 only

The numeric columns have reasonable ranges (for example satisfaction score from 1 to 10 and monthly spending from about 10 to 200), so no rows were removed or changed.

7. Methodology

- Loaded the CSV and checked shape, data types, missing values and duplicates.
- Calculated the KPIs directly from the data.

- Created account age groups using 12-month bands (0–12, 13–24, 25–36, 37–48, 49–60 months). The highest account age in the data is 59 months.
- Created spending groups from monthly spending: 0–50 Low, 50–100 Medium, 100–150 High, 150–200 Very High.
- Calculated churn rate for each group as churned customers divided by customers in the group.
- Calculated Pearson correlation between each numeric column and Churn.
- Built the dashboard so every number is recalculated in the browser from the CSV when a filter is changed.

The analysis is descriptive. Significance tests were not carried out, so differences between groups are described as observed and not as proven.

8. Exploratory Data Analysis

Retained and churned customers look very similar on the numeric columns. The average age, account age, spending, usage hours, support calls, late payments, streaming usage, discount usage, satisfaction score and complaint tickets differ only slightly between the two groups. For example, average monthly spending is 104.47 for retained customers and 105.55 for churned customers.

Gender, last interaction type and promo opt-in were also checked. Churn is 31.72% for female and 30.90% for male customers, and between 30.89% and 31.55% across the three interaction types. The differences are small.

9. KPI Analysis

KPI	Value
Total customers	8,000
Churned customers	2,505
Retained customers	5,495
Overall churn rate	31.31%
Average monthly spending	104.80
Average usage hours	254.33
Average satisfaction score	5.46

About one customer in three has churned. The two groups are not equal in size but both are large enough to compare.

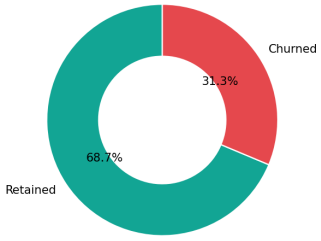


Figure 1: Retained vs churned customers

10. Churn Analysis

Figure 2 shows the churn rate for six customer breakdowns. The dashed red line is the overall churn rate (31.31%). The axis starts at zero so that the size of the differences is not exaggerated.

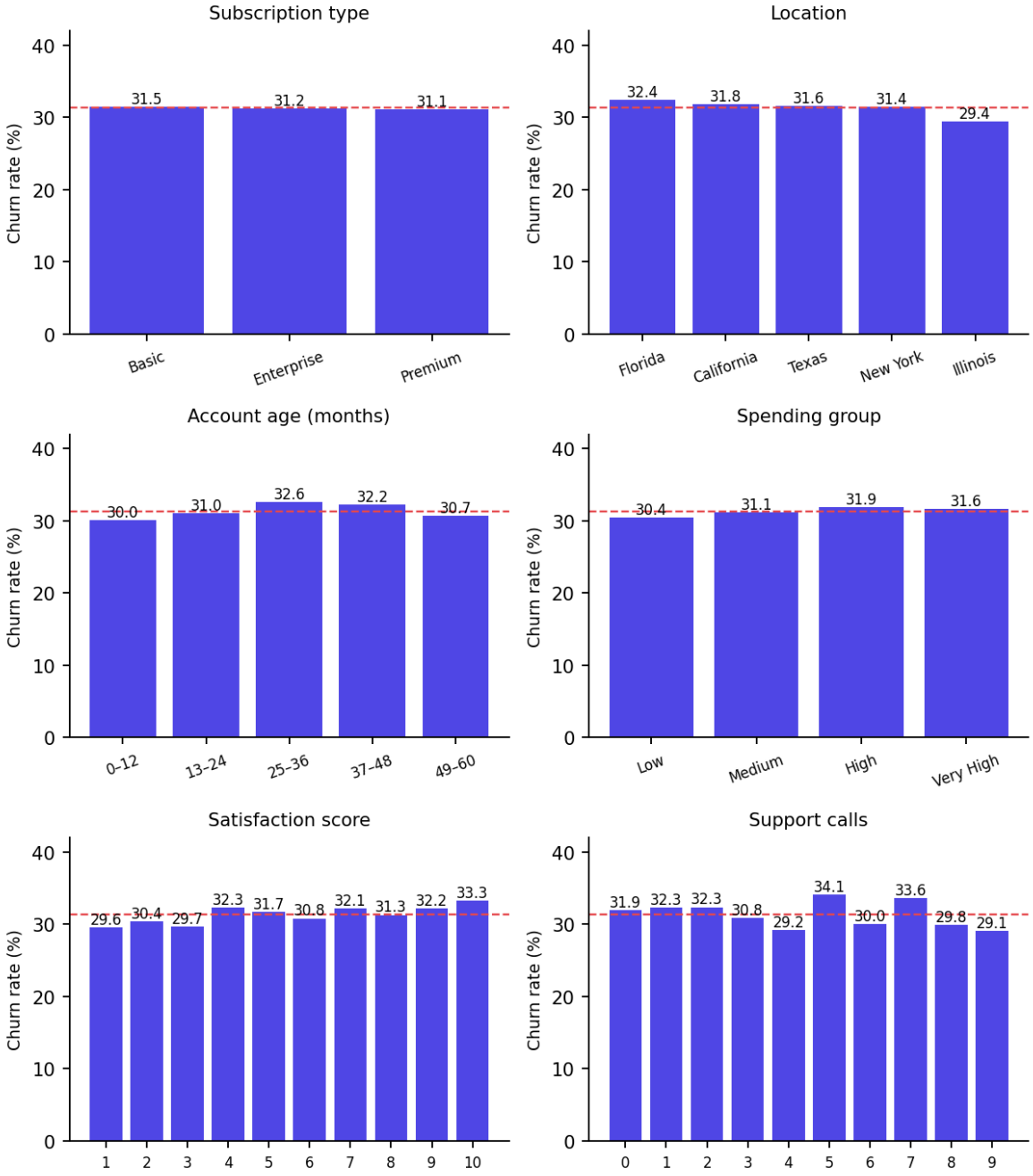


Figure 2: Churn rate (%) by segment

Subscription type

Basic is at 31.50%, Enterprise at 31.21% and Premium at 31.09%. The spread is under half a percentage point.

Location

Florida has the highest observed churn (32.37%), followed by California (31.81%), Texas (31.57%) and New York (31.44%). Illinois is the lowest at 29.43%.

Account age

Churn is 30.03% for 0–12 months, 31.03% for 13–24, 32.58% for 25–36, 32.18% for 37–48 and 30.65% for 49–60 months. The 25–36 and 37–48 month groups are a little higher, but the pattern is a small rise and fall and not a steady trend.

Spending

Low 30.45%, Medium 31.15%, High 31.88% and Very High 31.59%. The higher spending groups are slightly higher, but the gap between the lowest and highest is only about 1.4 percentage points.

Satisfaction score

Churn does not fall steadily as satisfaction rises. Score 1 has 29.58% and score 10 has 33.29%, and the scores in between move up and down. This should not be read as satisfaction affecting churn.

Support calls

Rates range from 29.09% (9 calls) to 34.10% (5 calls) with no steady direction. Each call count has roughly 770 to 870 customers, so part of this variation may simply be chance.

Correlation with churn

The correlations are all very weak (Figure 3). The largest are Discount_Used (0.0204) and Satisfaction_Score (0.0198); the smallest in size is Customer_ID (-0.0002), which is only an identifier. Late_Payments (-0.0166) and Promo_Opted_In (-0.0172) are slightly negative. These values should not be treated as evidence of causation.

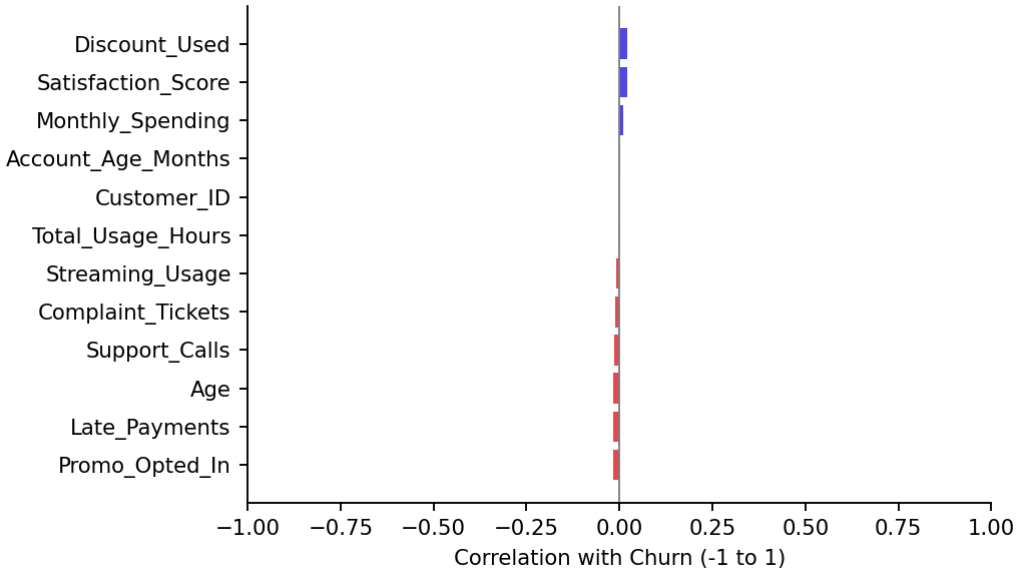


Figure 3: Correlation of numeric columns with Churn

Spending vs usage

Figure 4 plots monthly spending against total usage hours. Churned and retained customers are mixed together and do not form separate groups.

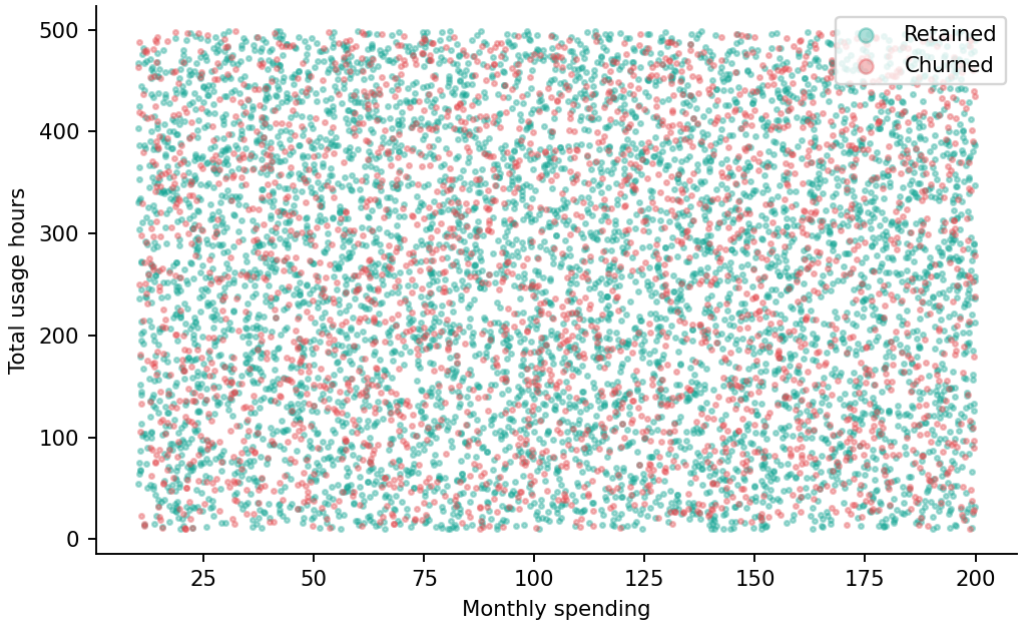


Figure 4: Monthly spending vs total usage hours

11. Key Findings

- Overall churn is 31.31%, about 1 in 3 customers.
- Florida has the highest observed location-level churn (32.37%) and Illinois the lowest (29.43%).
- Accounts aged 25–36 months (32.58%) and 37–48 months (32.18%) show relatively higher churn.
- Subscription types are almost identical in churn (31.09% to 31.50%).
- Spending groups differ by only about 1.4 percentage points.
- Satisfaction score and support calls show uneven patterns with no steady direction.
- All correlations with churn are between about -0.02 and +0.02.
- There is no strong standalone churn driver in this dataset. Churn is spread broadly, so looking at several factors together would be more useful than relying on one.

12. Business Recommendations

These suggestions follow from the observed patterns. They are starting points for investigation and not proven solutions.

1. Focus on mid-lifecycle customers. Accounts aged 25 to 48 months show slightly higher churn. A targeted retention initiative for this group could be tested.

2. Investigate geographic differences. Florida is higher than Illinois. Pricing, service quality, competition and customer behaviour could be compared between the two before concluding why.

3. Investigate support interactions. Some support-call counts show higher churn. The reason for each call and how it was resolved should be studied first.

4. Segment retention campaigns. Use combinations of location, subscription, account age, spending and engagement instead of one strategy for everyone.

5. Do deeper analysis. See Future Scope.

13. Dashboard

The dashboard is a single HTML page that reads train.csv in the browser and draws the charts with Plotly.js. It contains KPI cards, the charts described above, a correlation chart, insights and recommendations that are generated from the filtered data, and a searchable customer table. Nine filters (gender, location, subscription type, account age group, spending group, satisfaction score, churn status, promo opt-in and last interaction type) and a reset button update every number and chart without reloading the page. If no customer matches the filters, a message is shown instead.

The screenshots below were taken from the working dashboard with the full dataset loaded and no filters applied.

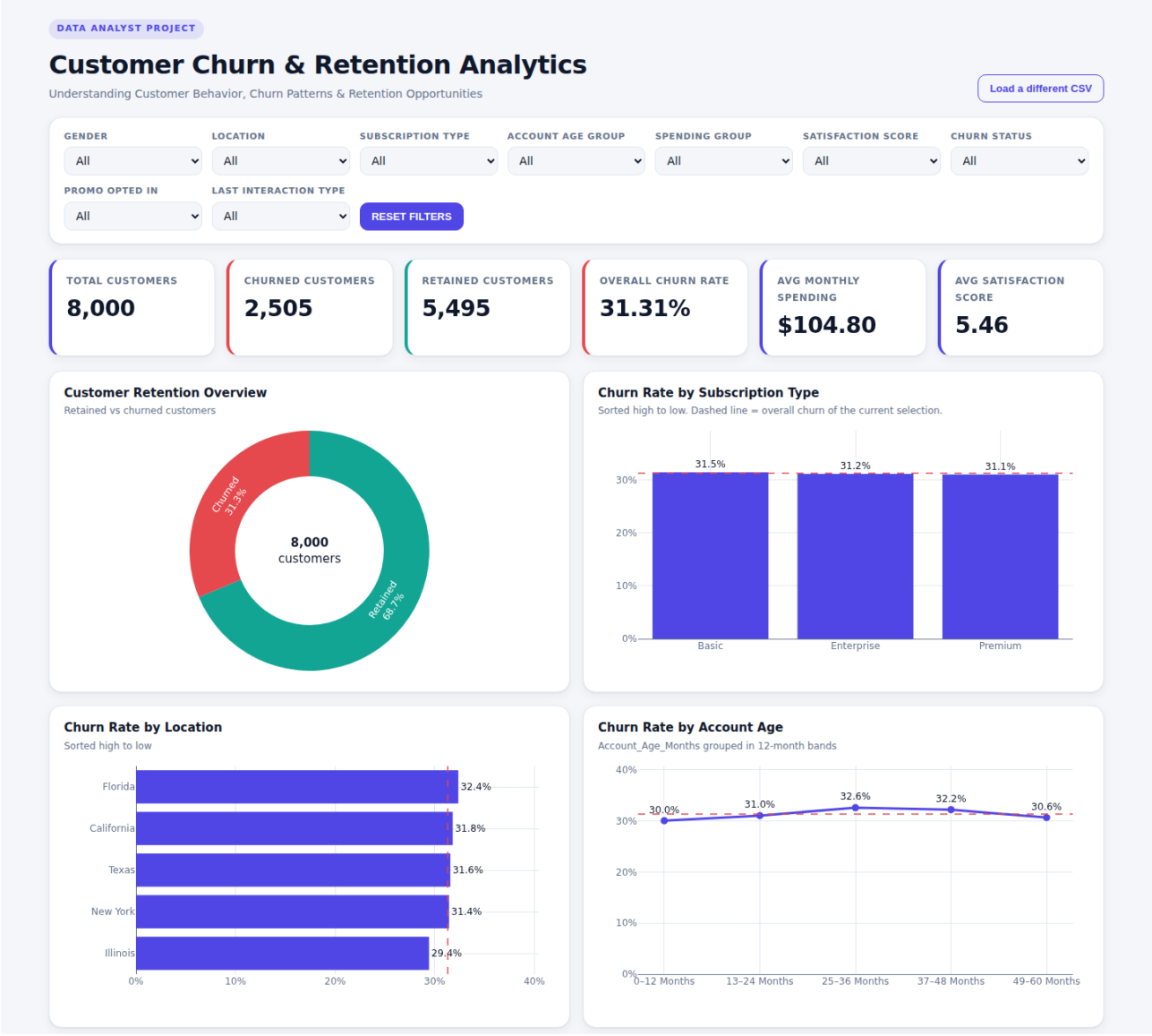


Figure 5: Dashboard header, filters, KPI cards and first charts

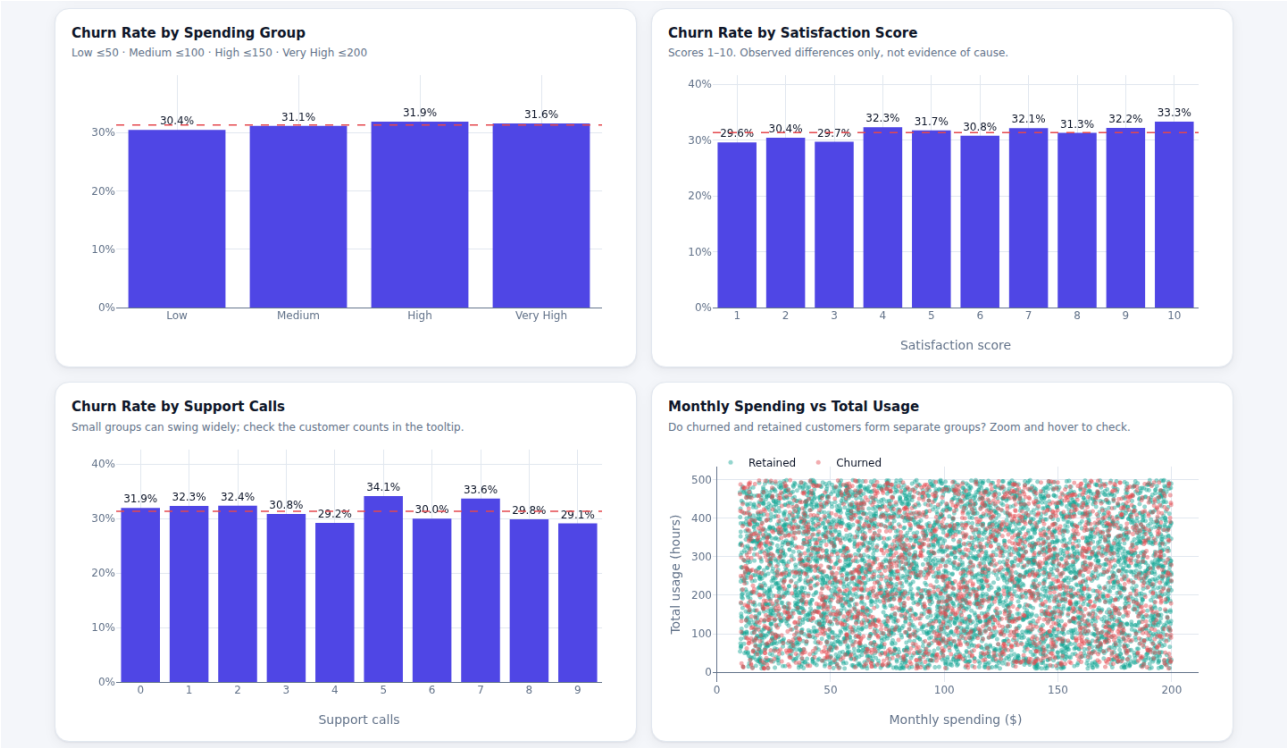


Figure 6: Spending, satisfaction, support-call and scatter charts



Figure 7: Correlation chart, key business insights and retention recommendations

14. Limitations

- The dataset is a single snapshot without dates, so trends over time cannot be studied.
- The reasons why customers left are not recorded.
- No significance testing was done, so some of the small differences may be due to chance.
- Single support-call counts and satisfaction scores are small groups (roughly 770 to 870 customers per support-call count).
- Variables were mostly examined one at a time; combined effects were not modelled.
- The original source of the dataset is not documented.

15. Future Scope

The following have not been done in this project and would be logical next steps:

- Customer segmentation, for example clustering
- Classification models for predicting churn
- Feature importance
- Cohort analysis
- Survival analysis
- Statistical tests for the group differences

16. Conclusion

Of 8,000 customers, 2,505 churned (31.31%). Churn rates by subscription, location, account age, spending, satisfaction and support calls all fall in a narrow range and the correlations with churn are close to zero. The data therefore does not show a strong standalone churn driver. Some segments are slightly higher than average and are worth further investigation, but these are small observed differences and not evidence of cause. A multi-factor, segment-based approach would be more suitable than a single-variable explanation.